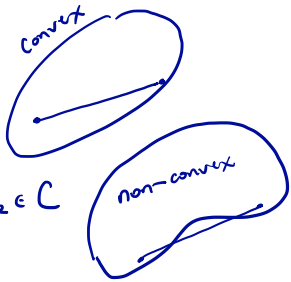
Convex Sets

Basic definition



$C \subseteq \mathbb{R}^n$, is convex if

$x_1, x_2 \in C$, $\lambda \in [0,1] \Rightarrow \lambda x_1 + (1-\lambda) x_2 \in C$

$\forall x_1, x_2 \in C$, $\forall \lambda \in [0,1]$,

$x = \lambda x_1 + (1-\lambda) x_2 \in C$.

Convex Sets - Examples

Positive semidefinite matrices : A $n \times n$ is symmetric if $A = A^T$: $A \in S^n$.

Using Algebra

$A \in S^n$ is psd if all eigenvalues of A are nonnegative : $A \in S^n_+$.

$\iff A \in S^n$ is psd if $x^T A x \geq 0 \ \forall x \in \mathbb{R}^n$.

$A_1, A_2 \in S^n_+$, $\lambda \in [0,1]$, $x^T(\lambda A_1 + (1-\lambda) A_2) x = \lambda \underbrace{x^T A_1 x}_{\geq 0} + (1-\lambda) \underbrace{x^T A_2 x}_{\geq 0} \geq 0 \ \forall x$

Using intersection of convex sets

$C_1, C_2, \ldots, C_K$ convex $\Rightarrow \bigcap C_i$ is convex.

$S^n_+ = \{A \in S^n : x^T A x \geq 0 \ \forall x\} = \bigcap_x \underbrace{\{A \in S^n : x^T A x \geq 0\}}_{S^n_x}$

Now need to show: S_x is convex.

Show that $S_x = \{A \in S^n,\ x^T A x \geq 0\}$

is convex.

Matrix Algebra: $x^T A x = \underbrace{\sum_{i,j} x_i x_j A_{ij}} = X^T \mathrm{vec}(A)$

Recall: $u^T v = \sum u_i v_i$

A – imagine it is (or write it as...) a n^2 long vector whose entries are A_{ij}. $= \mathrm{vec}(A)$

X – n^2 vector whose entries are $x_i \cdot x_j$.

$x^T A x \geq 0$: a half-space: $H^+_{s,b} = \{x : s^T x \geq b\}$

$s = X$: $H^+ = \{A : \underset{s}{X^T} \mathrm{vec} A \geq \underset{b}{0}\}$.

Convex Sets - Examples

degree d

Polynomials lower bounded on a set $\rightarrow$ convex.

$$\mathcal{P}_{d,\ell} = \{ p(x) : p(x) \geq \ell \quad \forall x \in S \}.$$

$$p_1, p_2 \in \mathcal{P}_{d,\ell} \qquad \lambda p_1 + (1-\lambda) p_2 \overset{?}{\in} \mathcal{P}_{d,\ell}$$

$$\forall x \in S : \quad \underbrace{\lambda p_1(x)}_{\geq \ell} + (1-\lambda) \underbrace{p_2(x)}_{\geq \ell} \geq \lambda \ell + (1-\lambda)\ell = \ell$$

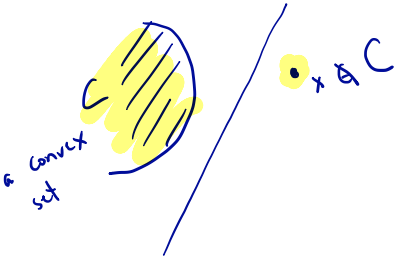
Polynomials whose max on a set is lower bounded

$$Q_{d,\ell} = \left\{ p(x) : \left[\max_{x \in S} : p(x) \right] \geq \ell \right\}$$

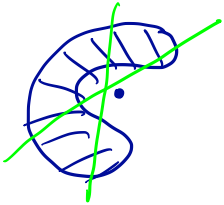
<u>Exercise</u>: show that $Q_{d,\ell}$ is <u>not</u> convex.

Separating Hyperplanes

The basic principle



Note: If C is not convex, this need not hold.



Separating Hyperplanes

An Example:

$$C = \mathbb{R}^n_+ \quad , \text{ say } n=3: \; C = \mathbb{R}^3_+$$

$$v = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \notin \mathbb{R}^3_+$$

The principle on prev. slide says there exists a hyperplane $H_{s,b} = \{x : s^T x = b\}$

We need to find: $s \in \mathbb{R}^3$, $b \in \mathbb{R}$ s.t.

$$\boxed{\begin{array}{l} s^T v < b \\ \text{and } s^T x \geq b \quad \forall x \in \mathbb{R}^3_+ \end{array}}$$

$$s = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad b = 0 \; : \quad s^T v = 0 \cdot 1 + 0 \cdot 2 + 1(-1) = -1 < b = 0$$

$$s^T x = 0 \cdot x_1 + 0 \cdot x_2 + 1 \cdot x_3 \geq b = 0.$$